Exp.No: 8a)

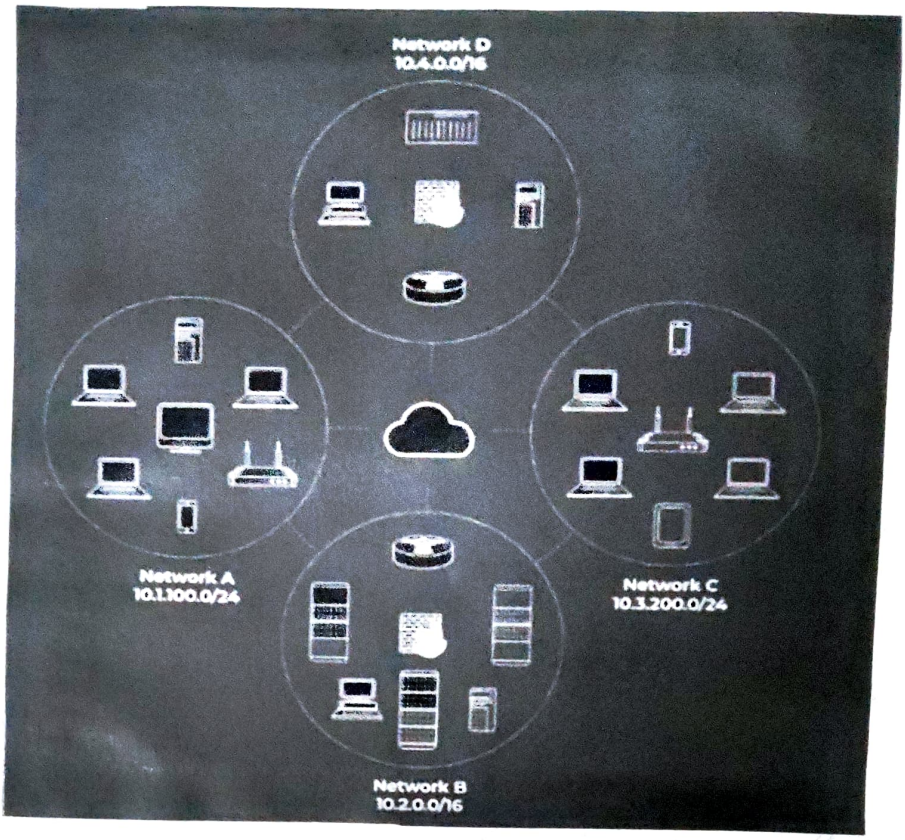
# NMap Like Host and Discovery using ARP, ICMP, TCP and UDP

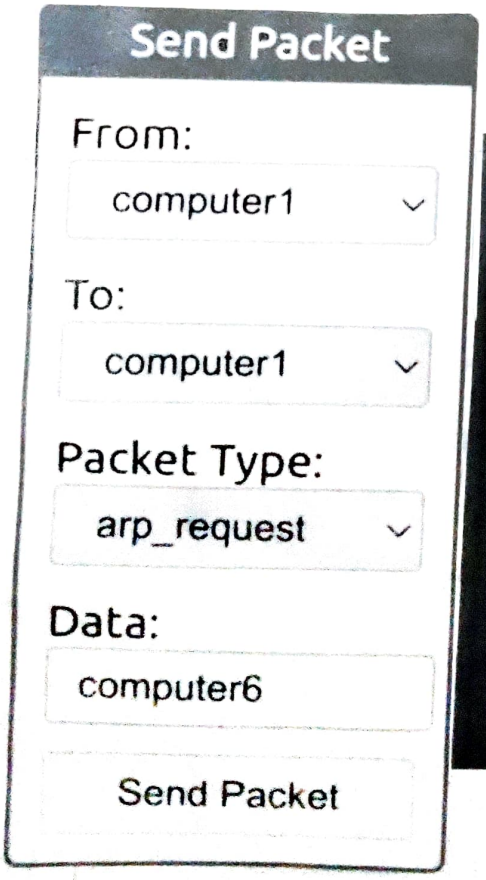
## Aim:

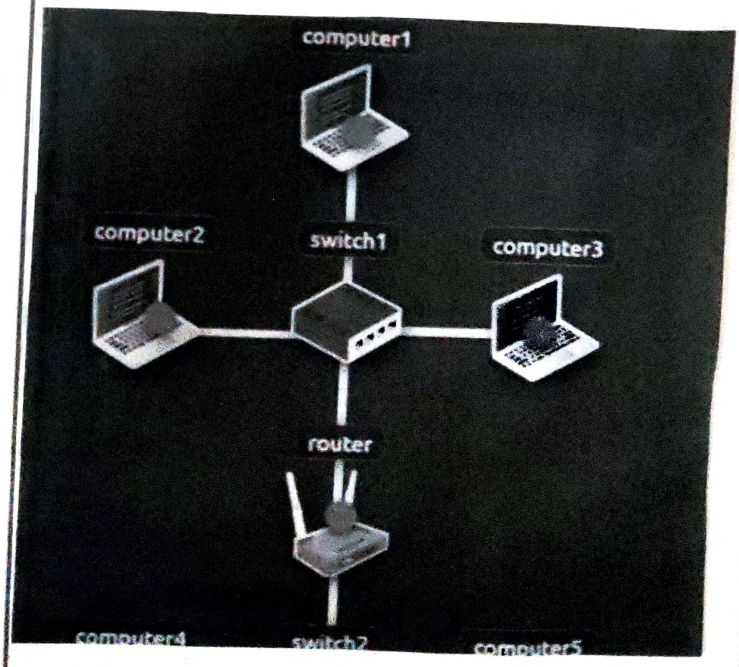
To discover like hosts using NMap Scans such as ARP, ICMP, TCP and UDP on the TryHackMe Platform.

## Algorithm:

1. Open the TryHackMe platform and access the NMap Like Host Discovery room.
2. Study the concept of like host discovery before performing network scanning.
3. Start with an ARP scan to identify like hosts in the local network.
4. Observe the ARP requests and responses from the target hosts.
5. Perform an ICMP scan to check whether the target systems respond to ICMP requests.
6. Perform an TCP SYN Ping Scan by sending TCP packets to the required ports.
7. Perform an TCP ACK ping Scan and observe the response from the target host.
8. Perform a UDP ping Scan to determine whether the target system is reachable.
9. Compare the results obtained from ARP, ICMP TCP and UDP host-discovery methods.
10. Record the like host identified by NMap and study the Scan results.


Network D
10.4.0.0/16
Network A
10.1.100.0/24
Network C
10.3.200.0/24
Network B
10.2.0.0/16


Send Packet
From:
computer1
To:
computer1
Packet Type:
arp_request
Data:
computer6
Send Packet


computer1
computer2
switch1
computer3
router
computer4
switch2
computer5

## Send Packet

From:

computer4

To:

computer4

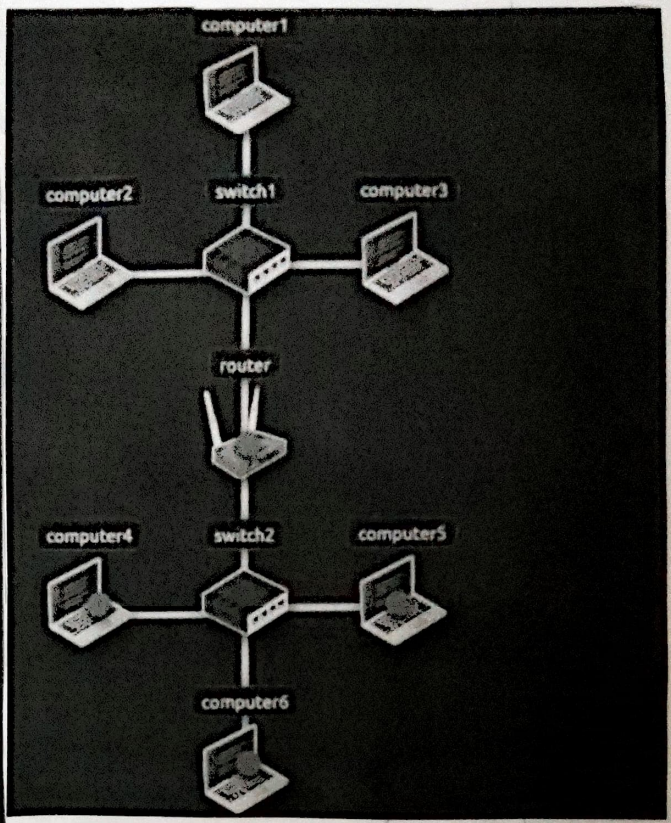
Packet Type:

arp_request

Data:

computer6

Send Packet




| | ISO/OSI | TCP/IP | |
|---|---|---|---|
| 7 | Application Layer | Application Layer | HTTP, HTTPS, SMTP, POP3, IMAP, SSH, FTP, SNMP, Telnet, RDP... |
| 6 | Presentation Layer | | |
| 5 | Session Layer | | |
| 4 | Transport Layer | Transport Layer | TCP, UDP |
| 3 | Network Layer | Network Layer | IPv4, IPv6, ICMP, IPsec |
| 2 | Data Link Layer | Link Layer | ARP, Ethernet (802.3), WiFi (802.11), DSL, Bluetooth, |
| 1 | Physical Layer | | |